

Temporal and linguistic properties of infant-parent communication in the home environment

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Introduction

Parents respond to infant communication in a variety of ways. Research suggests that the type of input and temporal contingency of the response lays critical groundwork for language development and is directly linked to later vocabulary development (Tamis-LeMonda, Bornstein, Kahana-Kalman, Baumwell, & Cyphers, 1998).

Infants change the quality (e.g., phonetic/phonological structure) of their vocalization based on the verbal response from the parent (Bloom et al., 1987; Goldstein & Schwade, 2008), suggesting that the parent response may shape aspects of the vocalization.

Our objective is to characterize the temporal and linguistic properties of infant-parent communication in the **home environment**.

Research Questions:

1. How often do infants vocalize?
2. How often parents respond?
3. Does the length of an infant vocalization affect parental response time?
4. When parents respond to the infant vocalization, how often do they label objects?

General Methods

Participants n = 3; (all 11 months, 2 M; 1F)

Home videos were acquired using a head camera sent home with families. Videos were only coded for **mealtime**, **playtime**, and **book reading scenes**, then parsed into 5-second segments.

Coding:

- Adult speech transcription
- Presence and duration of infant vocalizations
- Adult response time to the vocalization (within 6 sec., but longer responses accepted if it was clearly related to infant vocalization)

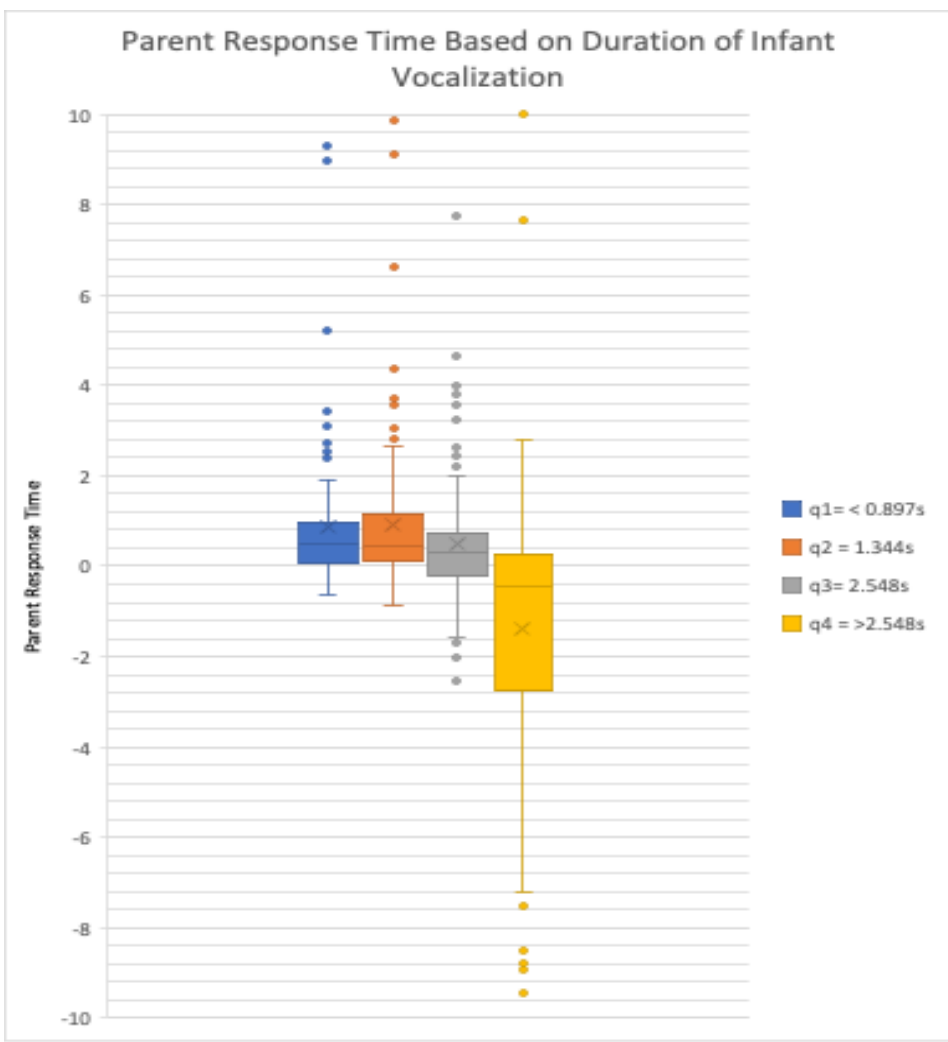


Question 1: How often do infants vocalize?

Methods: Created a datavyu column that contained every audible sound that the infant produced
-IV: Subject
-DV: Number of vocalizations produced

Question 2: How often do parents respond to infant vocalizations?

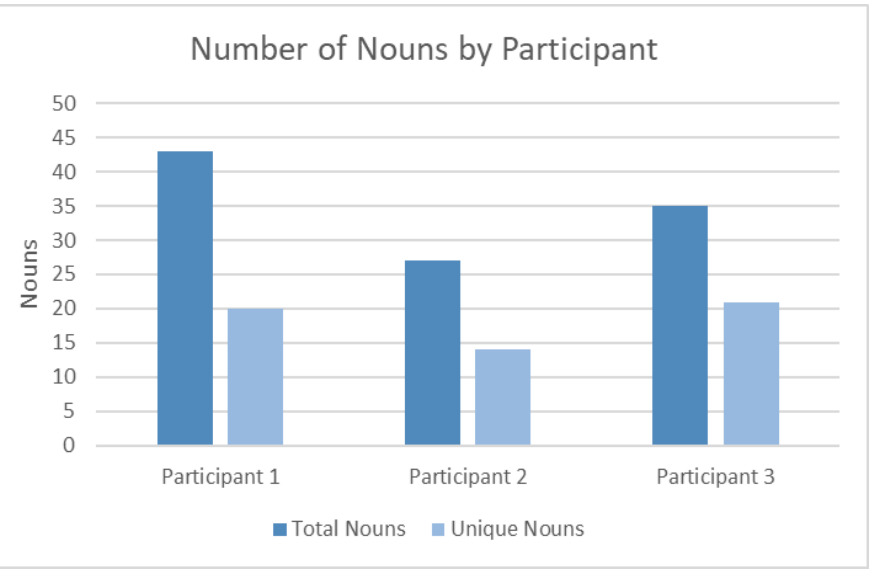
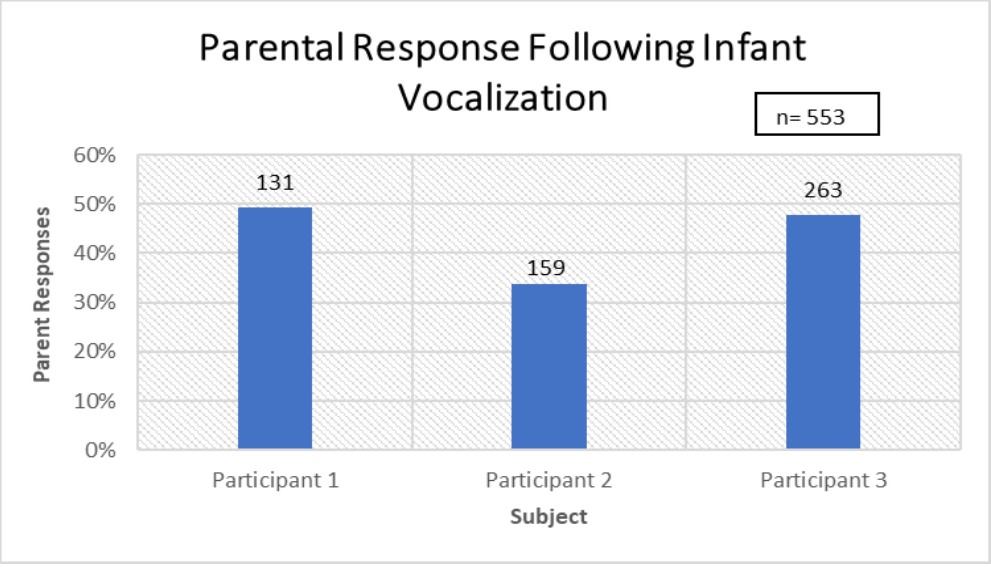
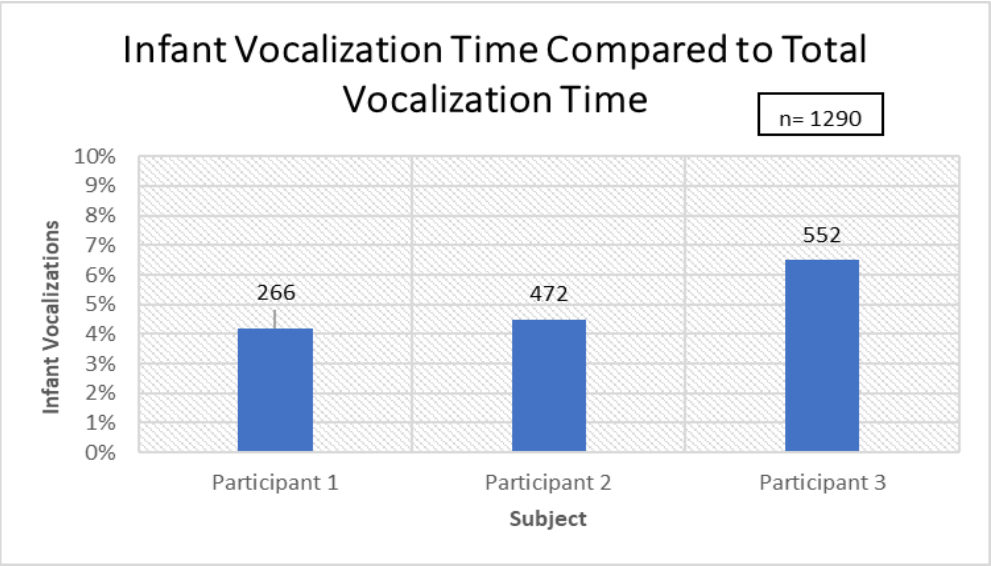
Methods: Created a datavyu column that contained every contextually relevant parental response to infant vocalization
-IV: Subject
-DV: Number of parental responses/non-responses



Question 3: Does the length of the infant’s vocalization affect the response time from the parents?

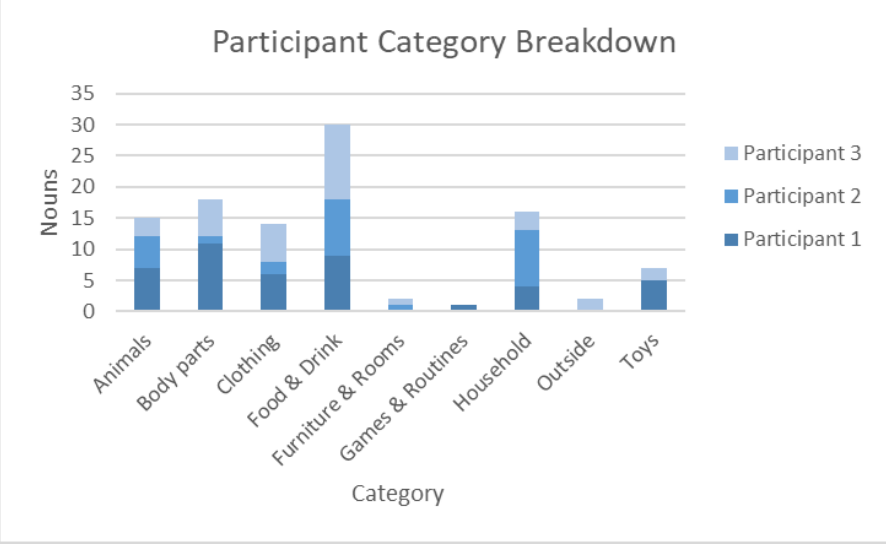
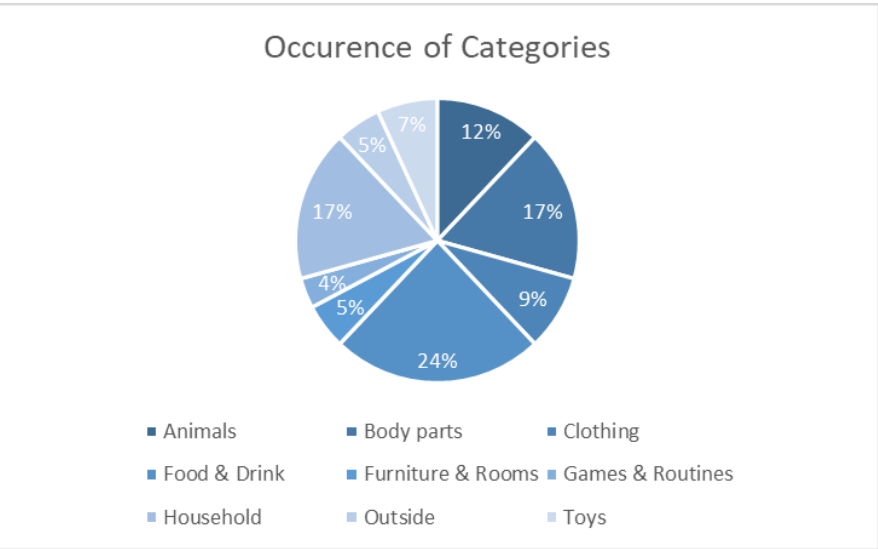
Methods: Used datavyu columns and onset and offset tools to measure the duration of vocalizations and the start of the response
-IV: The length of the vocalization (grouped)
-DV: The time it takes the parent to respond (response time)

	<0.897 sec	>.897 and <1.344 sec	> 1.344 & <2.548 sec	> 2.548 & < 18.227 sec
Mean	0.9178	0.9293	0.4399	-1.700
Median	0.4825	0.444	0.31	-0.465
SD	1.73357109	1.70969403	1.34198398	3.84334976



Question 4: What is the content of parental response? What noun labeling or noun usage occurs in a response?

Methods: Used datavyu to code for nouns and noun categories on the MCDI (vocabulary checklist) in the adult responses
-IV: Subject
-DV: Nouns and noun categories used during response



Summary

Question 1:

- Infants vocalized 5.06% of the time

Question 2:

- Parents responded to infant vocalizations 42.87% of the time

Question 3:

- Shorter response time as duration gets longer

Question 4:

- Parental responses contain a noun 18.98% of the time.
- Most labeling occurs in relation to food & drink. There’s significant repetition of words, but also diversity in the number of semantic categories infants hear.

Conclusion

Although infant vocalizations make up a minimal portion of home environment speech, it has a strong impact on non-infant conversation, eliciting parent response and noun labels

Limitations & Future Directions

- How rate of vocalization and parent response change as infants age
- Extending analyses to other home scenes beyond mealtime, playtime, and book reading
- Other phonetic/acoustic properties of the vocalization (e.g., pitch, loudness)
- Other aspects of infant communication (e.g., gestures, facial reactions) and parent response (e.g., touch, eye gaze)

References

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